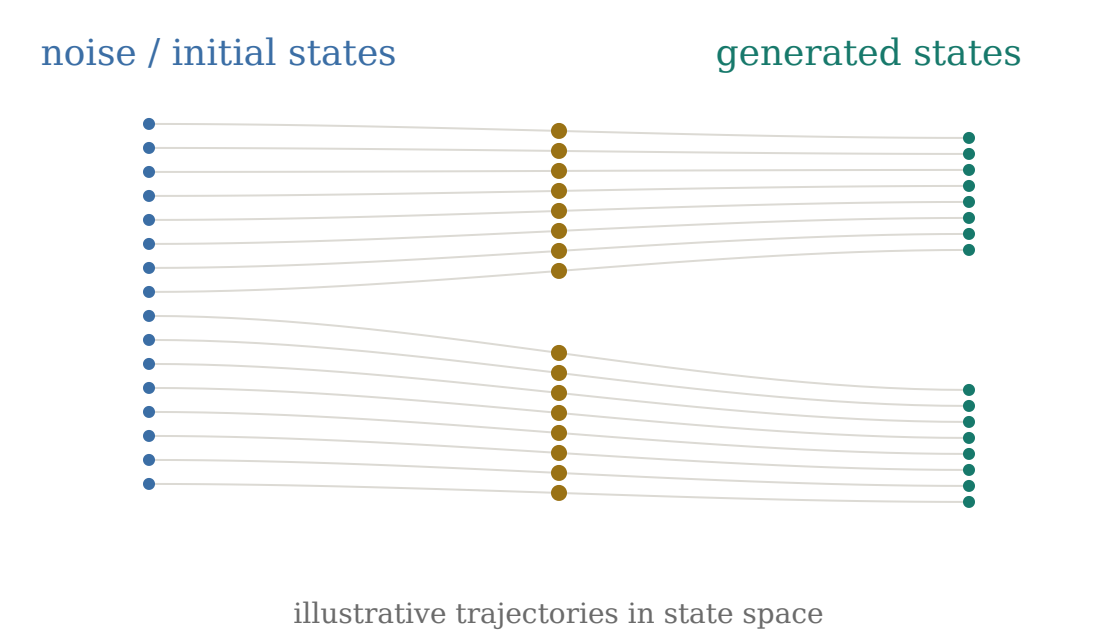
MOTIVATION

Flow Matching and VLA

From random vectors to robot actions

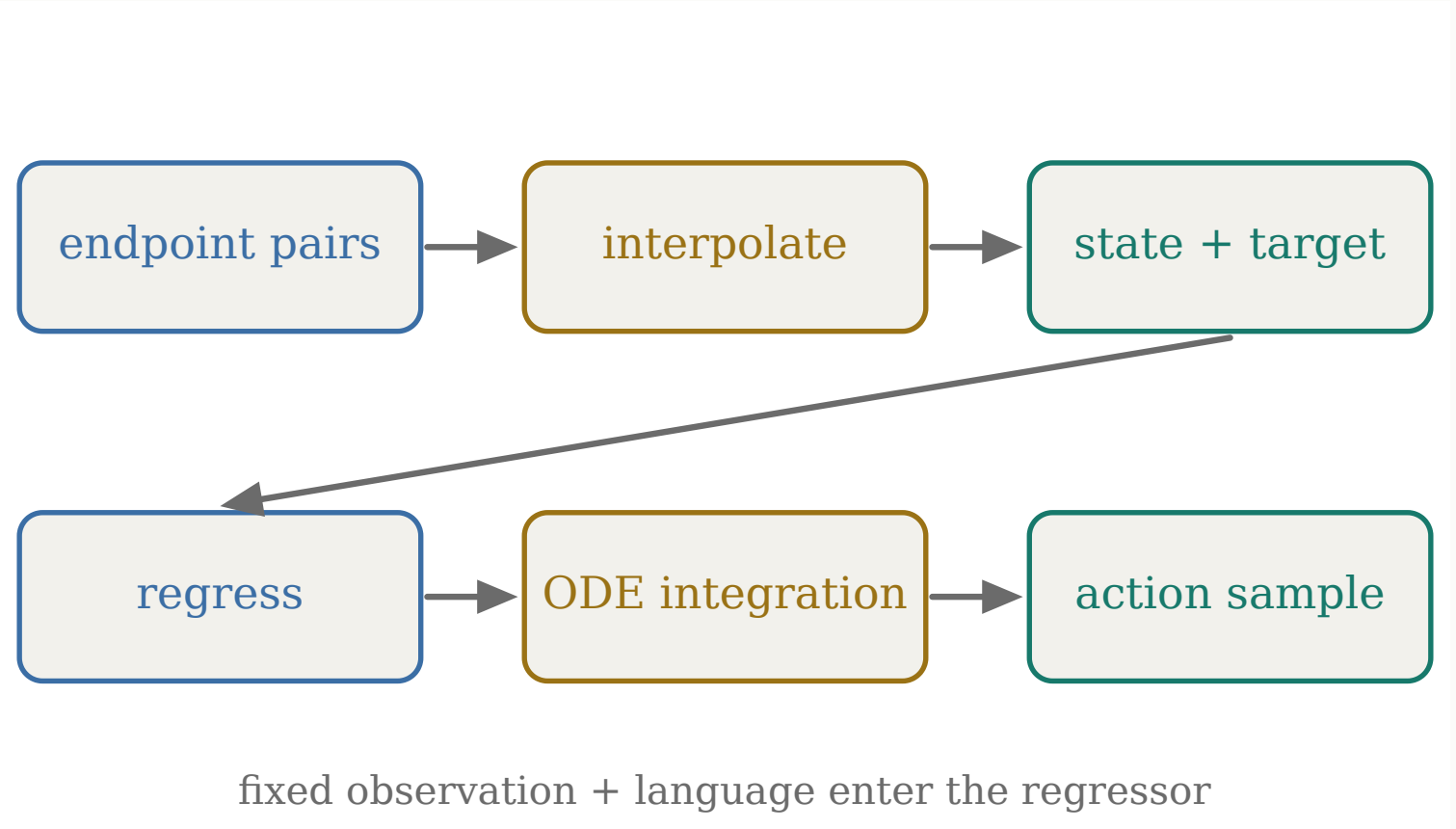


A vision-language-action (VLA) model generates actions from images and instructions.

ROADMAP

Flow Matching and VLA

The mathematical route



fixed observation + language enter the regressor

π_0, §IV and Appendix A-D

RECAP

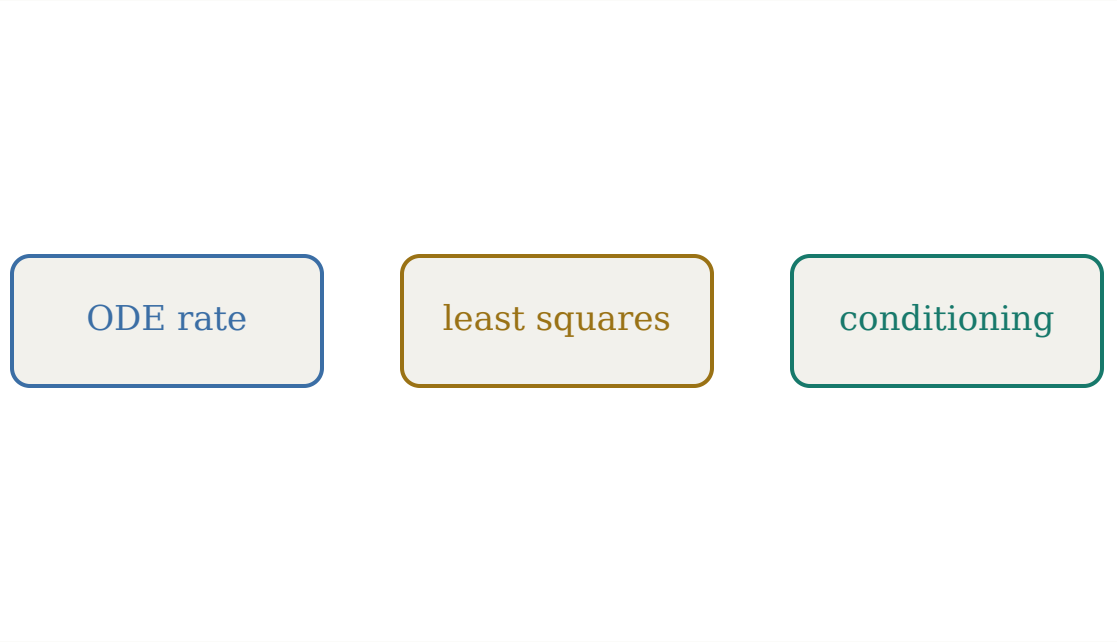
Flow Matching and VLA

Three results return

ODE: a rate determines a trajectory under existence and uniqueness conditions.

Squared error: conditional means minimize risk.

Conditioning: fix the observation when sampling.



Goodfellow et al., §3.8; current prerequisite links below

Recall: Lecture 7, p22: Euler · Lecture 11, pp5,34: conditional mean under squared loss · Lecture 8, pp35–36: conditioning

EXERCISE

Flow Matching and VLA

Which clock is running?

k: physical robot step; $\tau \in [0, 1]$: generator time. One action chunk is refined while the camera observation stays fixed.

A. Every flow step moves the robot.

B. Flow steps refine a candidate chunk.

C. Flow time is the camera frame index.

π_0, §IV and Appendix A-D

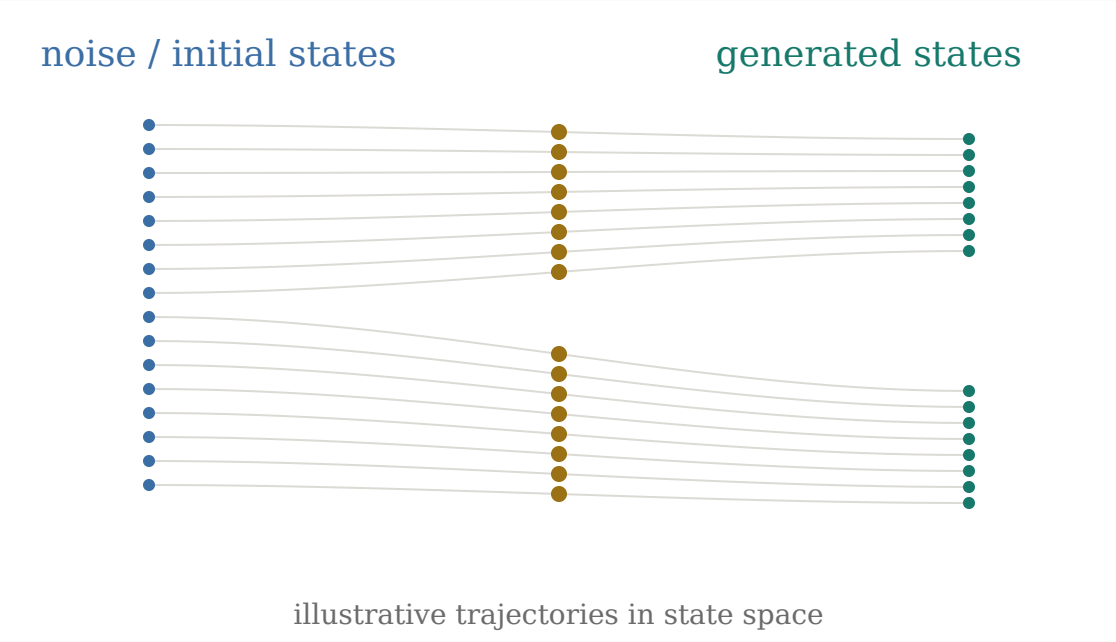
MOTIVATION

Probability Flows

The target is a distribution

$$X_0 \sim p_0, \qquad X_1 \sim p_{\text{data}}$$

where $X_0, X_1 \in \mathbb{R}^d$: random vectors with d coordinates. p_0: noise law; p_{data}: data law.



illustrative trajectories in state space

Lipman et al., §2, Eqs. (1)–(4)

DEFINITION

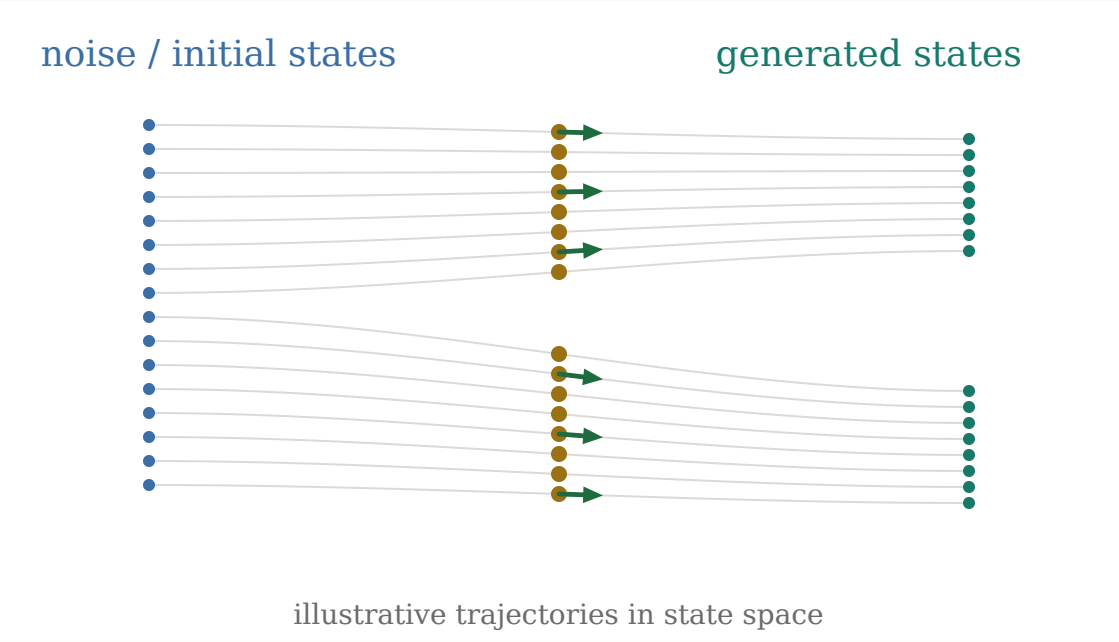
Probability Flows

A learned velocity field

A velocity field assigns a rate vector to each state and generation time.

$$\frac{dx_\tau}{d\tau} = v_\theta(x_\tau, \tau), \quad x_0 \sim p_0$$

where $x_\tau \in \mathbb{R}^d$: candidate at time $\tau \in [0,1]$. v_θ: learned rate; x_0: initial draw from p_0; θ: fixed weights.



illustrative trajectories in state space

Tong et al., §2.1, Eqs. (1)–(2), footnote 1

DEFINITION

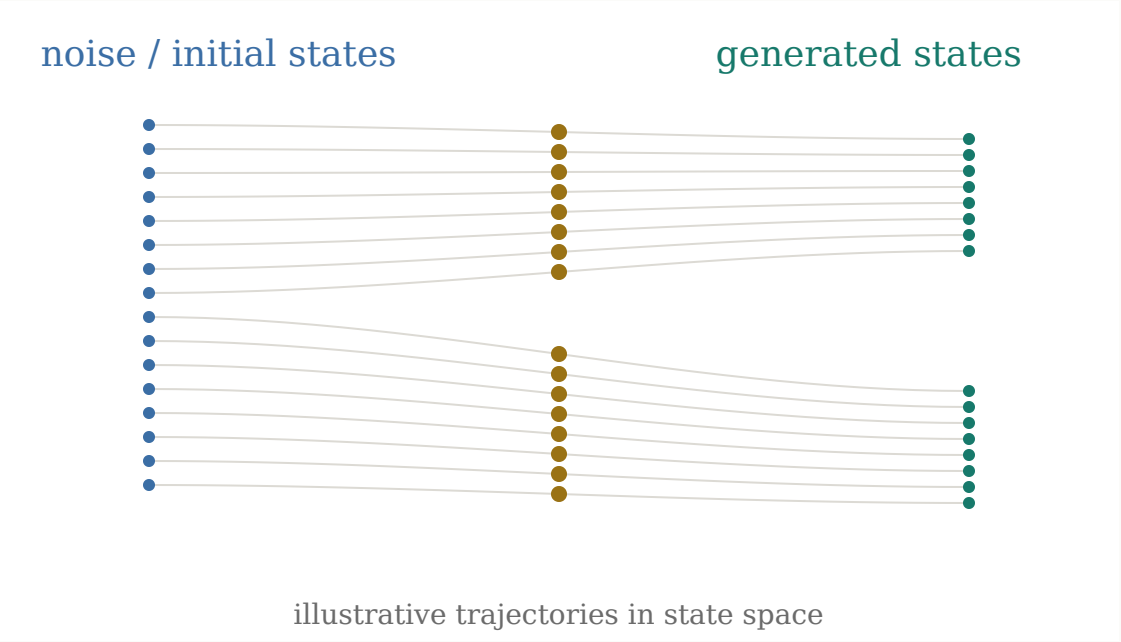
Probability Flows

The flow map collects trajectories

A flow map returns the state reached from an initial point after integrating the velocity field.

$$\phi_\tau(x_0) = x_\tau, \qquad \phi_0(x_0) = x_0$$

where x_0: initial point; x_τ: point at time τ. $\phi_\tau : \mathbb{R}^d \to \mathbb{R}^d$ on the domain of a well-posed flow.



illustrative trajectories in state space

[Tong et al., §2.1, Eqs. (1)–(2), footnote 1]

EXAMPLE

Probability Flows

Worked example: translation

$$v(x, \tau) = 2, \quad \phi_\tau(x_0) = x_0 + 2\tau$$

$$x_\tau - x_0 = \int_0^\tau 2 \, ds = 2\tau$$

$$\dot{x}_\tau = 2, \quad \phi_0(x_0) = x_0; \quad X_\tau \sim \mathcal{N}(2\tau, 1)$$

where x_0, x_τ: scalar samples; τ: generation time. Start with $X_0 \sim \mathcal{N}(0, 1)$, mean 0 and variance 1.

Lipman et al., §2, Eqs. (1)–(4)

EXERCISE

Probability Flows

Transfer: a different velocity

The initial law is $X_0 \sim \mathcal{N}(1, 4)$ (variance 4). Each state x_τ has velocity $dx_\tau/d\tau = -3$. Find the law at generation time $\tau = 1$.

A Gaussian mean -2, variance 4

B Gaussian mean -2, variance 16

C Gaussian mean 4, variance 4

Lipman et al., §2, Eqs. (1)–(4)

DERIVATION

Probability Flows

Expansion changes density

$$x_\tau = e^{a\tau} x_0, \qquad p_\tau(x) = e^{-a\tau} p_0(e^{-a\tau} x)$$

$$\dot{x} = ax \;\Rightarrow\; x = e^{a\tau} x_0, \quad x_0 = e^{-a\tau} x$$

$$p_\tau(x)\, dx = p_0(e^{-a\tau} x) e^{-a\tau}\, dx$$

where a: fixed rate; τ: generation time; x_0: initial state; x: evaluation point. p_0, p_τ: initial and current densities.

Lipman et al., §2, Eqs. (1)–(4)

EXAMPLE

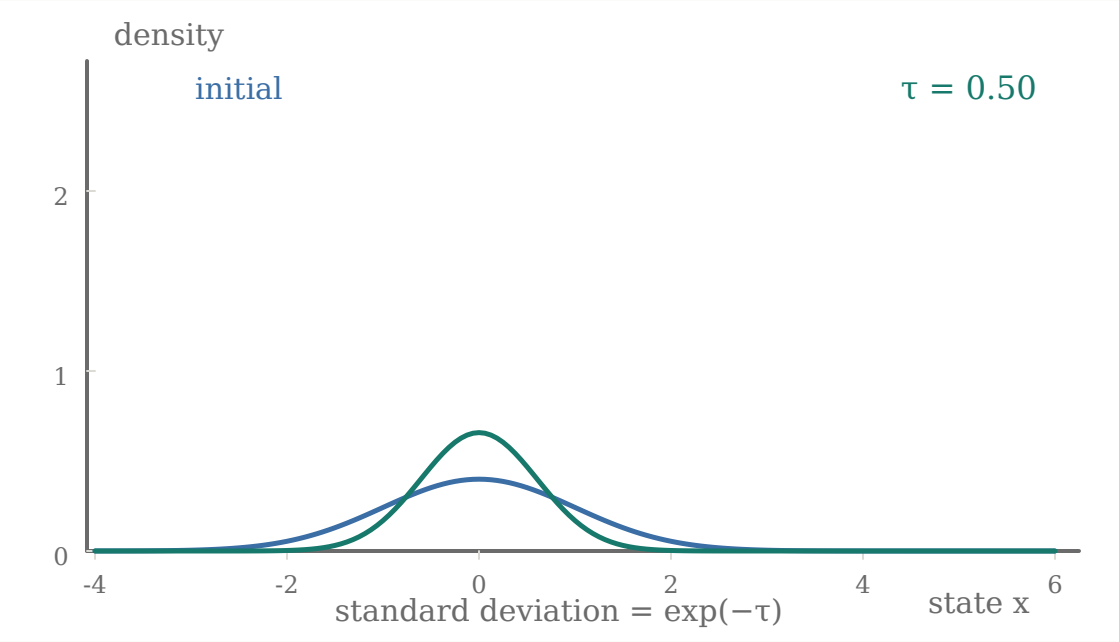
Probability Flows

Compression preserves total probability

Standard deviation: $e^{-\tau}$. Variance: $e^{-2\tau}$. Total area: 1.

$$X_\tau = e^{-\tau} X_0, \quad X_0 \sim \mathcal{N}(0, 1)$$

where X_0: initial random state; X_τ: state at time τ. $\mathcal{N}(0,1)$ has mean 0 and variance 1.



Lipman et al., §2, Eqs. (1)–(4)

DERIVATION

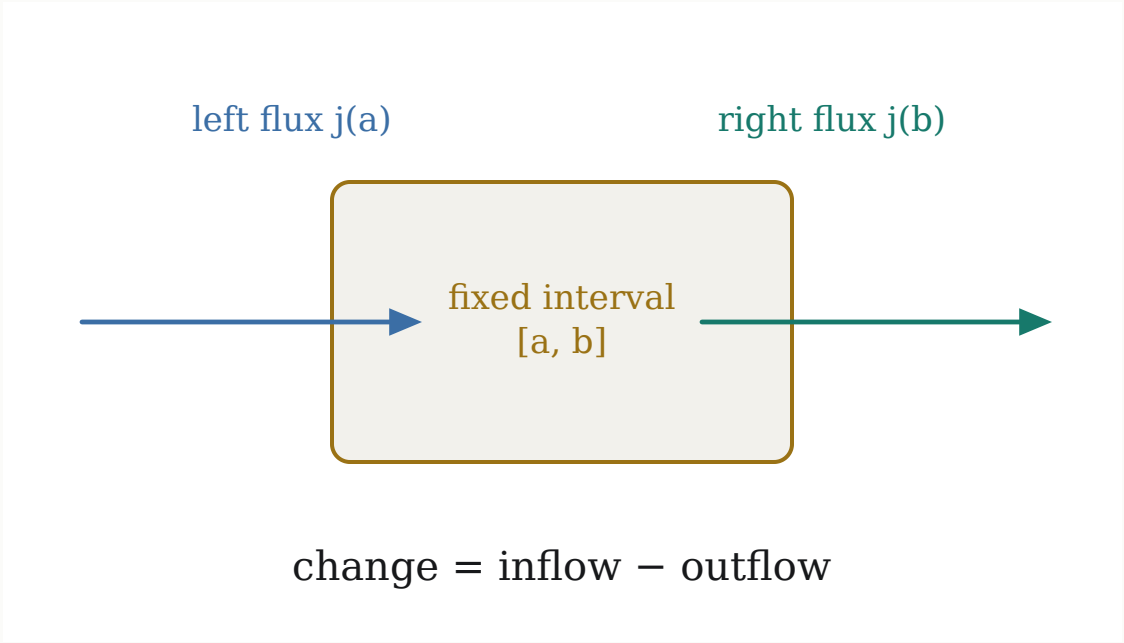
Probability Flows

Probability enters and leaves an interval

Probability flux is the signed probability crossing a boundary per unit generation time.

$$\frac{d}{d\tau}\int_a^b p_\tau(x)\,dx = j_\tau(a) - j_\tau(b)$$

where $[a, b]$: fixed interval; $p_\tau(x)$: density at x and time τ; $v(x, \tau)$: signed rate; $j_\tau = p_\tau v$: flux.



Lipman et al., Appendix B, Eq. (26)

DERIVATION

Probability Flows

Local probability conservation

$$\partial_\tau \int_a^b p\,dx = j(a) - j(b)$$

$$j(a) - j(b) = -\int_a^b \partial_x j\,dx$$

$$\int_a^b \left[\partial_\tau p + \partial_x (pv)\right] dx = 0$$

$$\partial_\tau p = -\partial_x j,\ \partial_\tau p + \nabla \cdot j = 0$$

where x, τ: position/time; p, v: smooth density/rate; $j = pv$: flux.
Fixed $[a, b]$; $\nabla\cdot$: divergence.

Lipman et al., Appendix B, Eq. (26)

EXERCISE

Probability Flows

Does constant velocity freeze density?

A smooth starting density p_0 translates at rate $v = 2$: $p_\tau(x) = p_0(x - 2\tau)$. At a fixed position x, does density stay constant as τ changes?

A. Yes, because velocity is constant.

B. No, because the density profile moves.

C. No, because total probability grows.

Lipman et al., Appendix B, Eq. (26)

EXAMPLE

Probability Flows

Integrated example: a moving Gaussian

$$X_\tau = 2\tau + (1+\tau)X_0, \quad X_0 \sim \mathcal{N}(0,1)$$

$$\dot{x}_\tau = 2 + x_0, \qquad x_0 = \frac{x_\tau - 2\tau}{1+\tau}$$

$$v(x,\tau) = 2 + \frac{x - 2\tau}{1+\tau}$$

$$X_1 = 2 + 2X_0 \sim \mathcal{N}(2,4)$$

where X_0, X_τ: initial/current random states; x_0, x_τ: sample values; τ: time. Scale $1+\tau > 0$.

Lipman et al., §2; Goodfellow et al., §3.8

EXERCISE

Probability Flows

Transfer: contracting toward a target mean

Initial $X_0 \sim \mathcal{N}(0, 4)$ has variance 4. The state at generation time τ is $X_\tau = 3\tau + (1 - \tau/2)X_0$. Find mean and variance at $\tau = 1$.

A Mean 3, variance 1

B Mean 3, variance 2

C Mean 0, variance 1

Lipman et al., §2; Goodfellow et al., §3.8

MOTIVATION

Conditional Paths

A training pair supplies both endpoints

A coupling is a joint law for two endpoints with specified individual distributions.

$$(X_0, X_1) \sim \pi, \qquad \pi_0 = p_0,\ \pi_1 = p_{\text{data}}$$

where X_0: noise endpoint; X_1: data endpoint. π: joint law; its marginals π_0, π_1 give the individual laws p_0, p_{data}.

Tong et al., §3.2.2

DEFINITION

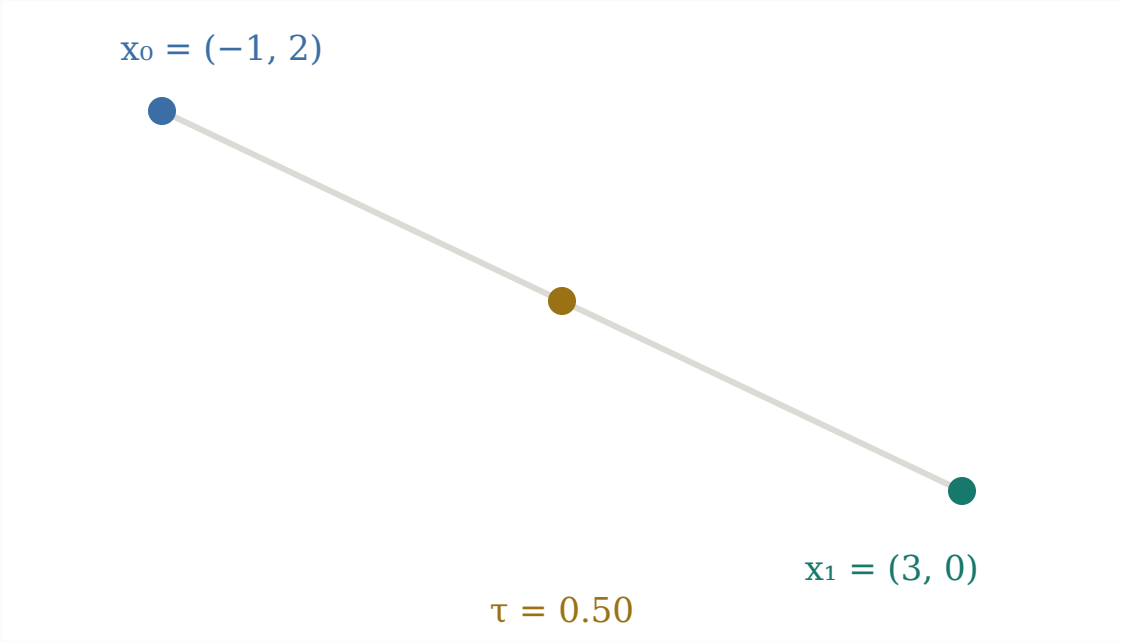
Conditional Paths

Straight interpolation defines a path

Linear interpolation forms a weighted average of two fixed endpoints.

$$X_\tau = (1-\tau)X_0 + \tau X_1, \quad 0 \le \tau \le 1$$

where $X_0, X_1 \in \mathbb{R}^d$: paired endpoints; τ: scalar interpolation weight.



Tong et al., §3.2.2

DERIVATION

Conditional Paths

The target velocity is an endpoint difference

$$U = \frac{dX_\tau}{d\tau} = -X_0 + X_1$$

$$\frac{d}{d\tau}[(1-\tau)x_0] = -x_0, \qquad \frac{d}{d\tau}[\tau x_1] = x_1$$

$$u = x_1 - x_0, \qquad \Delta x = \Delta\tau\, u$$

where X_0, X_1: fixed endpoints for this derivative; X_τ: intermediate state; τ: generation time; $U \in \mathbb{R}^d$: target rate.

Tong et al., §3.2.2

EXAMPLE

Conditional Paths

Worked example: a 2D pair

$$x_0 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad x_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad \tau = \tfrac{1}{4}$$

$$u = (3 - (-1),\, 0 - 2)^T = (4, -2)^T$$

$$x_{1/4} = (-1, 2)^T + \tfrac{1}{4}(4, -2)^T = (0, 3/2)^T$$

$$x_{1/4} + \tfrac{3}{4} u = (3, 0)^T = x_1$$

where x_0: noise; x_1: data; x_τ: state at generation time τ; u: target rate.

Tong et al., §3.2.2

EXERCISE

Conditional Paths

Transfer: another pair

Noise $x_0 = (2, -1)^T$, data $x_1 = (-2, 3)^T$, generation time $\tau = 1/4$. Choose the intermediate state x_τ and target velocity u.

A ((1, 0), (−4, 4))

B ((−1, 2), (−4, 4))

C ((1, 0), (−1, 1))

Tong et al., §3.2.2

DEFINITION

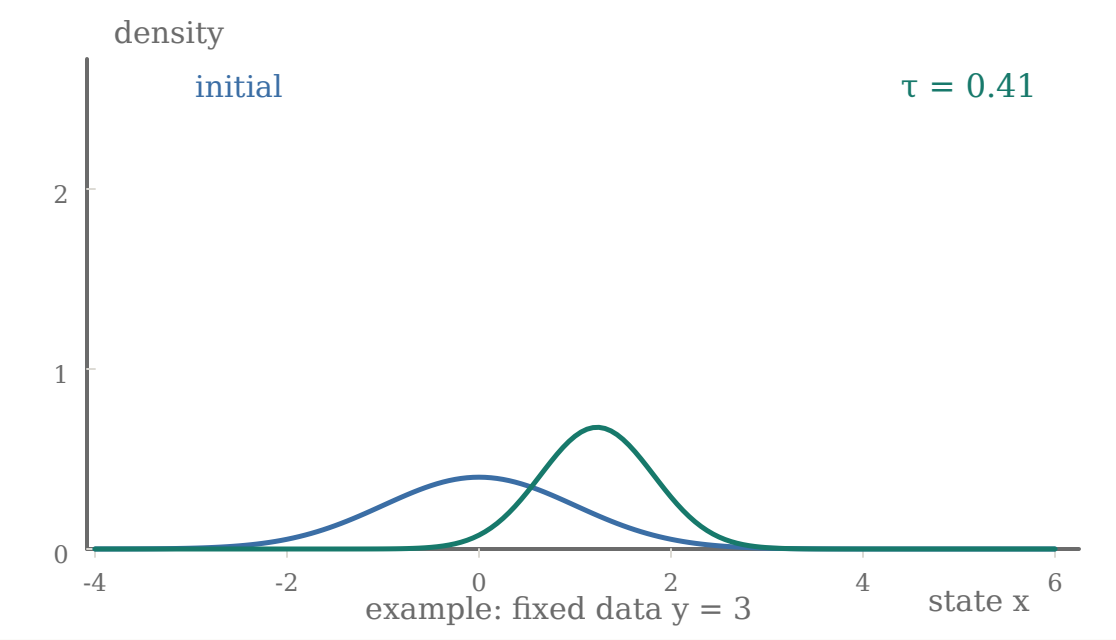
Conditional Paths

Fixing data leaves a Gaussian noise cloud

A conditional path describes the random intermediate state after fixing a data endpoint.

$$X_\tau \mid X_1 = y \sim \mathcal{N}(\tau y, (1-\tau)^2 I)$$

where $X_0 \sim \mathcal{N}(0, I)$ independent of X_1; I: identity covariance. Fixed $y \in \mathbb{R}^d$; $0 \le \tau < 1$.



Lipman et al., §4, Eqs. (10)–(14)

EXERCISE

Conditional Paths

Is the noise coefficient a variance?

Noise $X_0 \sim \mathcal{N}(0, 1)$, fixed data $y = 4$. The state is $X_\tau = (1 - \tau)X_0 + \tau y$. Find its mean and variance at time $\tau = 1/2$.

A Mean 2, variance 1/2

B Mean 2, variance 1/4

C Mean 4, variance 1/4

Lipman et al., §4, Eqs. (10)–(14)

EXERCISE

Conditional Paths

Can we use the endpoint during sampling?

Training uses noise x_0, demonstration x_1 and generation time τ to form the candidate x_τ. A new sampling call has only noise and an observation.

A. Give the network the unknown x_1.

B. Use $x_1 - x_0$ directly at every step.

C. Predict velocity from x_τ, τ and observation.

Lipman et al., §4, Eqs. (10)–(14)

DERIVATION

Conditional Paths

A conditional field can eliminate the noise

$$u(x, \tau \mid y) = \frac{y - x}{1 - \tau}, \quad \tau < 1$$

$$x_0 = \frac{x - \tau y}{1 - \tau}$$

$$y - x_0 = \frac{(1 - \tau)y - x + \tau y}{1 - \tau} = \frac{y - x}{1 - \tau}$$

where x_0: initial noise; y: fixed data; $x = (1 - \tau)x_0 + \tau y$: state; $u = y - x_0$: rate.

Lipman et al., §4, Eqs. (10)–(14)

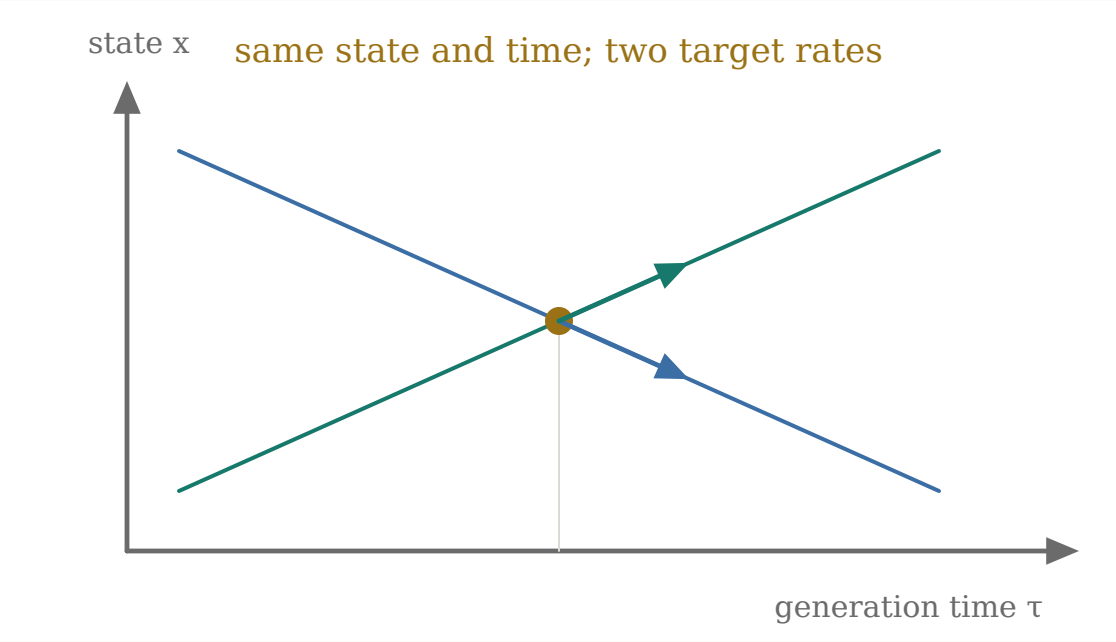
MOTIVATION

Conditional Paths

One input can have several target velocities

$$U = X_1 - X_0, \qquad X_\tau = x$$

where X_0, X_1: paired endpoints; X_τ: training state at time τ; x: shared location; U: random target rate.



Lipman et al., §3.1, Eq. (8)

EXERCISE

Conditional Paths

Which average does the field need?

At a fixed state x and generation time τ, target rates are $+2$ with probability $3/4$ and -2 with probability $1/4$. Which average should the model predict?

A. The unconditional endpoint average.

B. The target mean given this x and τ.

C. Always the most frequent target exactly.

Lipman et al., §3.1, Eq. (8)

DEFINITION

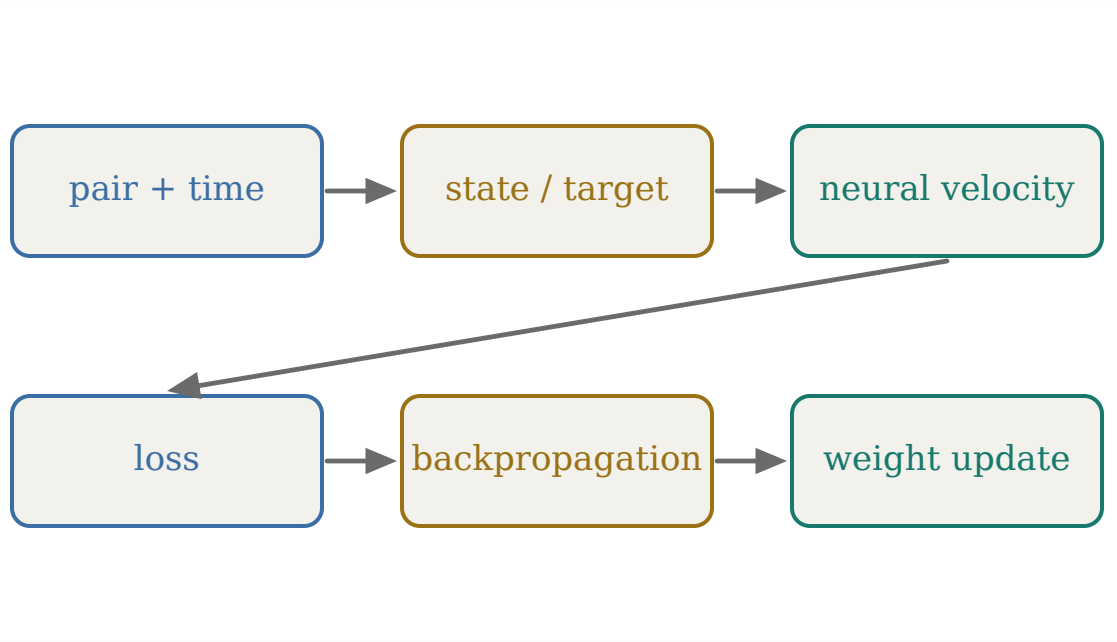
Learning the Velocity

Squared error compares rates

Conditional flow matching trains a velocity field against labels from sampled conditional paths.

$$\mathcal{L}(\theta) = \mathbb{E}\|v_\theta(X_\tau, \tau) - U\|^2$$

where X_τ: state at time τ; $U = X_1 - X_0$: target rate; v_θ: predicted rate; θ: weights.



Lipman et al., §3.2, Eq. (9)

DERIVATION

Learning the Velocity

Derivation: the best scalar prediction

$$R(a) = \mathbb{E}[(a - U)^2 \mid x, \tau]$$

$$R(a) = a^2 - 2a\mathbb{E}[U \mid x, \tau] + \mathbb{E}[U^2 \mid x, \tau]$$

$$R'(a) = 2a - 2\mathbb{E}[U \mid x, \tau]$$

$$a^* = \mathbb{E}[U \mid x, \tau], \qquad R''(a) = 2 > 0$$

where a: scalar prediction; U: target with finite second moment; x, τ : fixed input; R: risk.

Lipman et al., §3.2, Eq. (9)

RESULT

Learning the Velocity

The vector result is coordinatewise

A conditional mean averages only the targets compatible with the given input.

$$v^*(x, \tau) = \mathbb{E}[U \mid X_\tau = x, \tau]$$

$$\|a - U\|^2 = \sum_{j=1}^{d} (a_j - U_j)^2$$

$$\nabla_a R = 2(a - \mathbb{E}[U \mid x, \tau]) = 0$$

where U: vector target; a: prediction; R: risk at fixed state x and time τ. Finite second moments.

Lipman et al., §3.2, Eq. (9)

EXAMPLE

Learning the Velocity

Worked example: conflicting targets

$$R(a) = \tfrac{3}{4}(a-2)^2 + \tfrac{1}{4}(a+2)^2$$

$$R(a) = \tfrac{3}{4}(a^2 - 4a + 4) + \tfrac{1}{4}(a^2 + 4a + 4)$$

$$R(a) = a^2 - 2a + 4 = (a-1)^2 + 3$$

$$R(1) = 3 < 4 = R(2)$$

where a: predicted rate; R: conditional risk. Targets $2, -2$ have weights $3/4, 1/4$.

Lipman et al., §3.2, Eq. (9)

EXERCISE

Learning the Velocity

Transfer: changed conditional weights

At a fixed state x and generation time τ, the target U equals -1 with probability $1/4$ and 3 with probability $3/4$. Minimize conditional squared error.

A Predict 1

B Predict 2

C Predict 3

Lipman et al., §3.2, Eq. (9)

DERIVATION

Learning the Velocity

Conditional supervision learns the marginal field

$$\mathbb{E}\|v - U\|^2 = \mathbb{E}\|v - m\|^2 + \mathbb{E}\|U - m\|^2$$

$$v - U = (v - m) + (m - U)$$

$$\mathbb{E}[(v - m)^T (m - U) \mid X_\tau, \tau] = (v - m)^T 0 = 0$$

$$L_{\text{conditional}} = L_{\text{marginal}} + C, \qquad \nabla_\theta C = 0$$

where $m = \mathbb{E}[U|X_\tau, \tau]$: mean target; v: prediction; $C = \mathbb{E}\|U - m\|^2$.
Finite second moments.

Lipman et al., Theorem 2; course least-squares calculation

ALGORITHM

Learning the Velocity

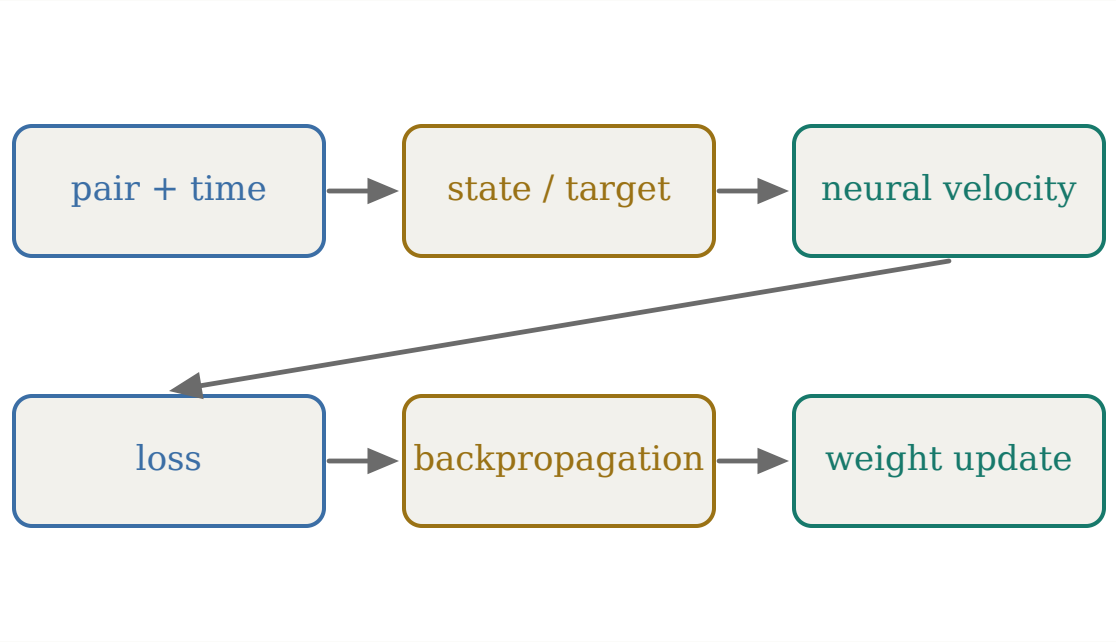
Training is ordinary supervised regression

Simulation-free training constructs velocity labels without solving model trajectories.

Sample endpoints and a generation time.

Construct the state and velocity target.

Predict velocity, compute squared error, backpropagate.



Lipman et al., Theorem 2; course least-squares calculation

EXERCISE

Learning the Velocity

Shapes and loss reduction

Prediction V and target U have $B = 2$ sample rows i and $d = 3$ coordinate columns j. Find the gradient of $L = \frac{1}{Bd}\sum_{i,j}(V_{ij} - U_{ij})^2$.

A. Gradient is 2(V−U)

B. Gradient is (V−U)/3

C. Gradient has shape 3×2

Lipman et al., Theorem 2; course least-squares calculation

EXAMPLE

Learning the Velocity

Worked example: a scalar linear field

$$v_w(x, \tau) = wx, \quad (x, u) = (1, 2), (2, 1)$$

$$L'(w) = (w - 2) + 2(2w - 1) = 5w - 4$$

$$w_0 = 0, \quad \eta = .1, \quad w_1 = 0 - .1(-4) = .4$$

$$L(0) = 2.5, \qquad L(.4) = 1.3$$

where w: weight; x, u: input, target; η: step size. $L(w) = [(w - 2)^2 + (2w - 1)^2]/2$.

Lipman et al., Theorem 2; course least-squares calculation

EXERCISE

Learning the Velocity

Transfer: a negative input

Rate $v_w(x) = wx$ with input $x = -2$, target $u = 3$, and weight $w = 1$. Take one descent step of size $\eta = .1$ on $L = \frac{1}{2}(wx - u)^2$.

A New w is 0

B New w is 2

C New w is 1.5

Lipman et al., Theorem 2; course least-squares calculation

RECAP

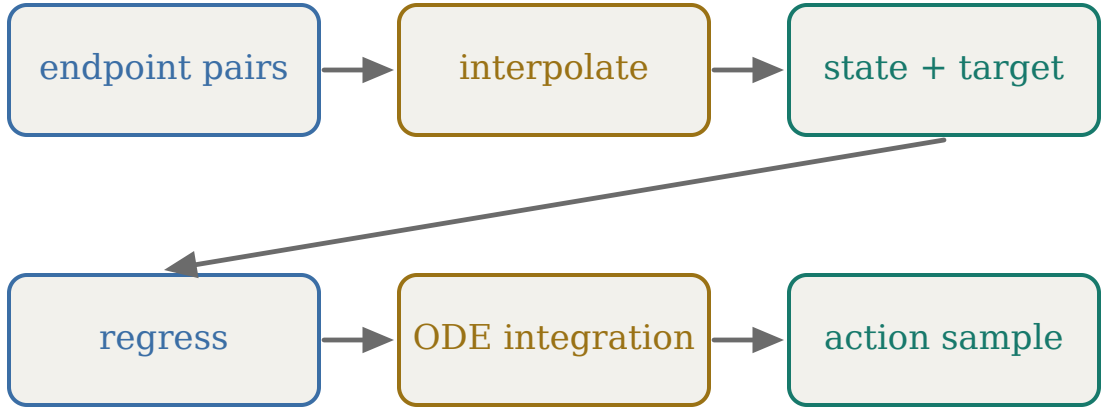
Learning the Velocity

The training chain is complete

Paired endpoints define targets.

Conditional least squares defines one field.

A new random start still needs integration.



fixed observation + language enter the regressor

[Lipman et al., Theorem 2; course least-squares calculation]

EXERCISE

Sampling the Learned Flow

What remains fixed while sampling?

Candidate x_τ follows $dx_\tau/d\tau = v_\theta(x_\tau, \tau, c)$. Draw x_0 once. τ: generation time; θ: field weights; c: fixed observation.

A. Optimize θ at every flow step.

B. Hold θ and c fixed; update x and τ.

C. Replace x with fresh noise every step.

Lecture 7, p22 and pp33–34; Lipman et al., §2

RECAP

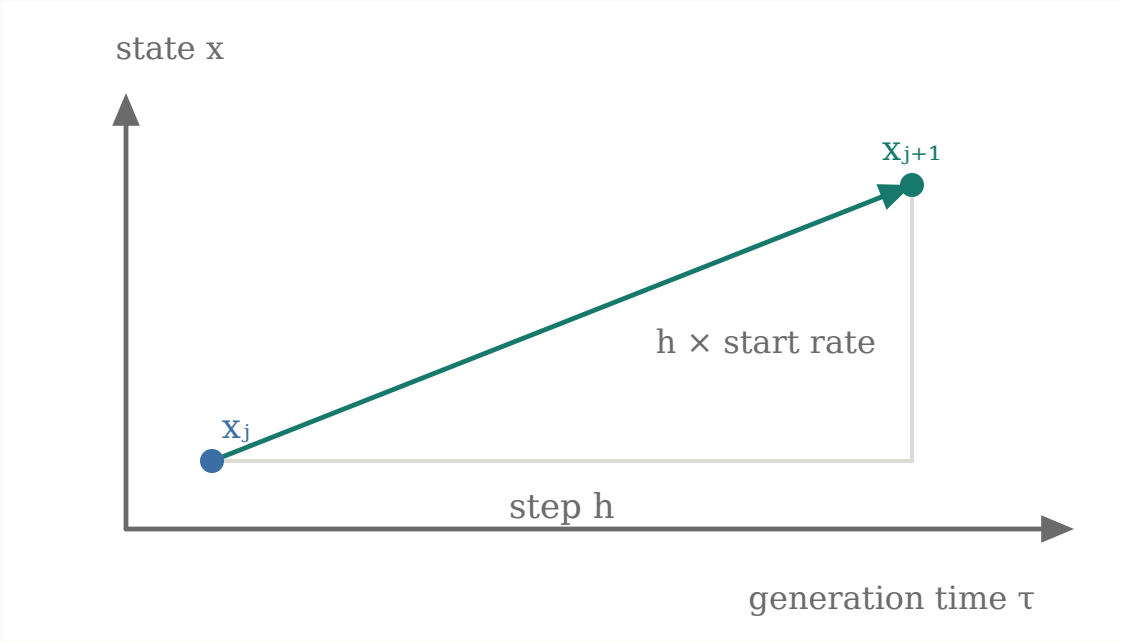
Sampling the Learned Flow

Euler multiplies velocity by a step size

Forward Euler advances a state using the rate at the beginning of the step.

$$x_{j+1} = x_j + h\, v_\theta(x_j, \tau_j, c), \quad \tau_{j+1} = \tau_j + h$$

where x_j: candidate at τ_j; j: solver index; c: fixed observation; θ : fixed weights. N steps of size $h = 1/N$.



Lecture 7, p22 and pp33–34; Lipman et al., §2

EXAMPLE

Sampling the Learned Flow

Worked example: two Euler steps

$$v(x, \tau) = x, \quad x_0 = 1, \quad h = \tfrac{1}{2}$$

$$x_1 = 1 + \tfrac{1}{2}(1) = 1.5$$

$$x_2 = 1.5 + \tfrac{1}{2}(1.5) = 2.25$$

$$x(1) = e \approx 2.71828$$

where x_j: numerical state after step j; h: step size; τ: time. Exact solution $x(\tau) = e^{\tau}$.

Lecture 7, p22 and pp33–34; Lipman et al., §2

EXERCISE

Sampling the Learned Flow

Transfer: a time-dependent rate

State x starts at $x_0 = 1$. The rate is $v(x, \tau) = 2\tau$. Use two forward Euler steps of size $h = 1/2$ from time $\tau = 0$ to 1.

A 1.5

B 2

C 2.5

Lecture 7, p22 and pp33–34; Lipman et al., §2

DERIVATION

Sampling the Learned Flow

A midpoint estimate improves the local rate

Explicit midpoint uses a trial half-step to estimate the rate for a full step.

$$k_1 = v(x_j, \tau_j), \quad x_m = x_j + hk_1/2$$

$$k_2 = v(x_m, \tau_j + h/2)$$

$$x_{j+1} = x_j + hk_2$$

where x_j, τ_j: start; h: full step; v: field for fixed context c; k_1, k_2: rates; x_m: trial state.

Lecture 7, p22 and pp33–34; Lipman et al., §2

EXAMPLE

Sampling the Learned Flow

Worked example: midpoint on the same field

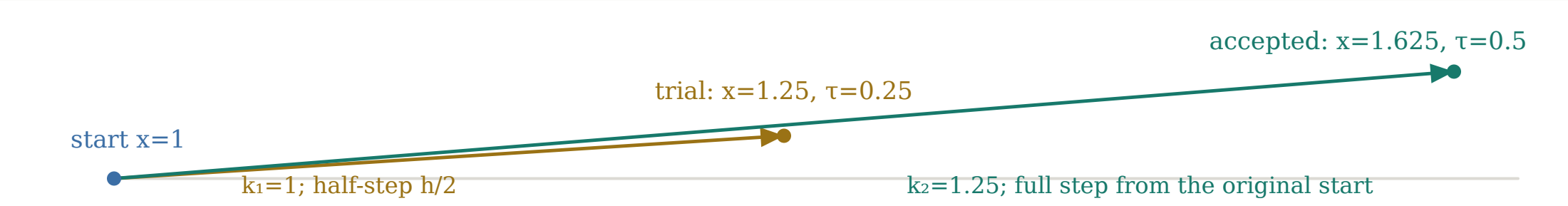
$$v(x,\tau) = x, \quad x_0 = 1, \quad h = \tfrac{1}{2}$$

$$k_1 = 1,\ x_m = 1 + .25(1) = 1.25,\ k_2 = 1.25$$

$$x_1 = 1 + .5(1.25) = 1.625$$

$$x_2 = (1 + h + h^2/2)^2 = 1.625^2 = 2.640625$$

where k_1, k_2: start/trial rates; x_m: trial state; h: full step. Accepted update $x_{j+1} = x_j + hk_2$.



Lecture 7, p22 and pp33–34; Lipman et al., §2

EXERCISE

Sampling the Learned Flow

Transfer: a fixed evaluation budget

Budget: 8 calls to learned rate v_θ. Euler uses 1 call/step; explicit midpoint uses 2. Both integrate $[0,1]$. Compare step sizes h.

A Euler h=1/8; midpoint h=1/4

B Both h=1/8

C Midpoint always gives the smaller error

Lecture 7, p22 and pp33–34; Lipman et al., §2

DIAGNOSTIC

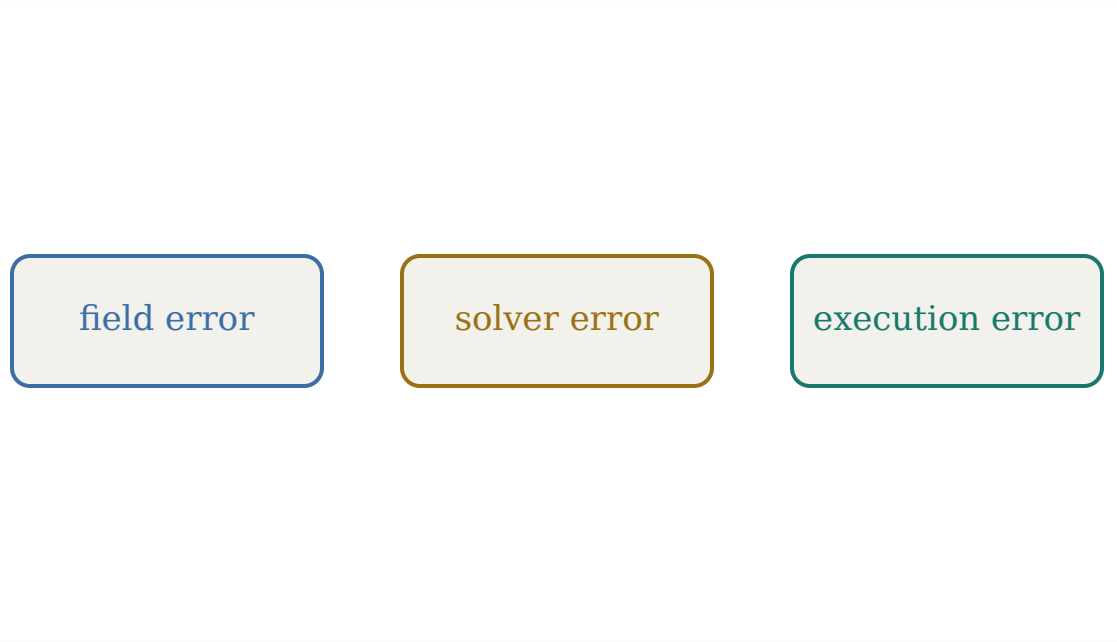
Sampling the Learned Flow

Different errors need different repairs

Field error: the learned velocity differs from the target.

Solver error: finite steps approximate that learned ODE.

Execution error: robot dynamics differ from assumptions.



Lecture 7, p22 and pp33–34; Lipman et al., §2

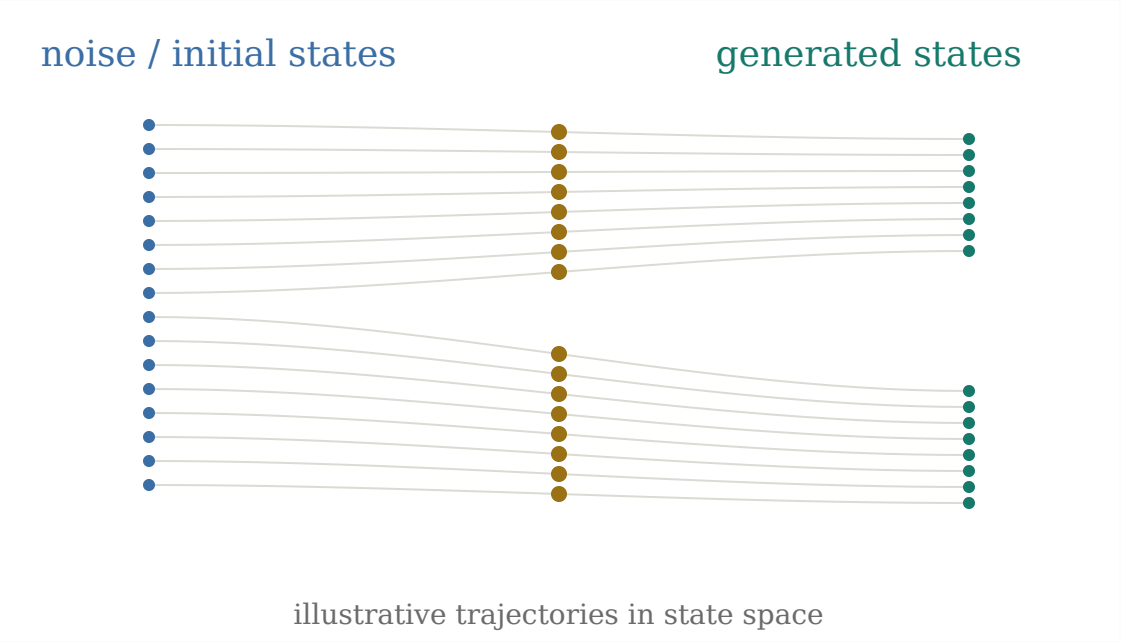
INTERPRETATION

Sampling the Learned Flow

Random starts preserve sample diversity

$$X_0 \sim p_0, \qquad X_1 = \phi_1(X_0; c)$$

where X_0, X_1: random start and endpoint; p_0: noise law; c: fixed observation; ϕ_1: deterministic endpoint map.



illustrative trajectories in state space

Lecture 7, p22 and pp33–34; Lipman et al., §2

EXERCISE

Sampling the Learned Flow

Reversing the generation clock

Course candidate x_τ has rate $v(x_\tau, \tau)$. Reverse the clock by $s = 1 - \tau$ and set $y_s = x_{1-s}$. Find the new rate dy_s/ds.

A $v(y_s, 1-s)$

B $-v(y_s, 1-s)$

C $-v(y_s, s)$ always

Lecture 7, p22 and pp33–34; Lipman et al., §2

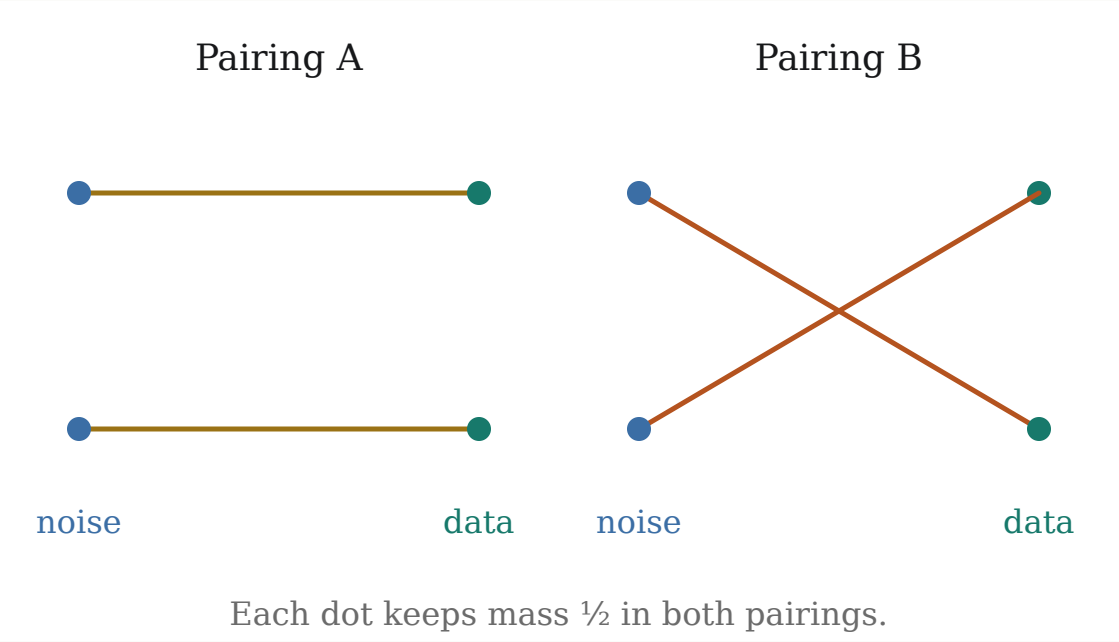
DEFINITION

Pairing and Optimal Transport

A coupling changes pairs, not endpoint laws

$$\pi(X_0 \in A) = p_0(A), \quad \pi(X_1 \in B) = p_1(B)$$

where X_0, X_1: noise/data endpoints; p_0, p_1: their laws; π: joint law; A, B: measurable sets.



Each dot keeps mass ½ in both pairings.

Tong et al., §§2.2, 3.2.3

EXAMPLE

Pairing and Optimal Transport

Worked example: two pairings

$$x_0 \in \{-1, 1\}, \quad x_1 \in \{-2, 2\}$$

$$C_{\text{aligned}} = \tfrac{1}{2}[(-2+1)^2 + (2-1)^2] = 1$$

$$C_{\text{crossed}} = \tfrac{1}{2}[(2+1)^2 + (-2-1)^2] = 9$$

where x_0, x_1: source/target values; each has mass $1/2$. C: mean squared displacement.

Tong et al., §§2.2, 3.2.3

DEFINITION

Pairing and Optimal Transport

Optimal transport chooses the coupling

Quadratic optimal transport minimizes expected squared displacement while preserving both endpoint laws.

$$\pi^* \in \arg\min_{\pi \in \Pi(p_0, p_1)} \mathbb{E}_\pi \|X_1 - X_0\|^2$$

where X_0, X_1: noise/data endpoints; p_0, p_1: their laws; Π: valid couplings; π^*: minimizing coupling.

Tong et al., §§2.2, 3.2.3

EXERCISE

Pairing and Optimal Transport

Transfer: compute the assignment cost

Sources $0, 2$; targets $1, 4$; each has mass $1/2$. Compare aligned $(0 \to 1, 2 \to 4)$ and crossed pairs.

A. Aligned cost 2.5; crossed 8.5

B. Aligned cost 5; crossed 17

C. Both cost 2.5

Tong et al., §§2.2, 3.2.3

EXERCISE

Pairing and Optimal Transport

Are learned trajectories necessarily straight?

Paired training paths are straight. At state x and time τ, $v^*(x,\tau) = \mathbb{E}[U|x,\tau]$ averages target rates U. Are generated paths straight?

A Yes, every generated path is one training segment.

B No; the conditional average can vary along a path.

C Every learned path must cross another path.

Lipman et al., §3.1, Eq. (8)

EXERCISE

Pairing and Optimal Transport

What does a conditional path establish?

Each conditional path moves Gaussian noise toward a fixed demonstration. Noise and demonstrations may be paired independently. What does this establish?

A. This alone proves globally optimal endpoint pairing.

B. A simple conditional path and global OT are different claims.

C. Every coupling has equal expected displacement cost.

Tong et al., §3.2.1, after Eq. (13)

EXERCISE

Pairing and Optimal Transport

Re-pair examples with different conditions

Condition c says “move left”. Its data actions are negative. A batch also contains “move right” examples.

A. Re-pair all targets freely and retain old c.

B. Keep targets compatible with each condition.

C. Remove c from training and still claim conditioning.

Tong et al., §§3.1–3.2; course conditional-pairing example

DEFINITION

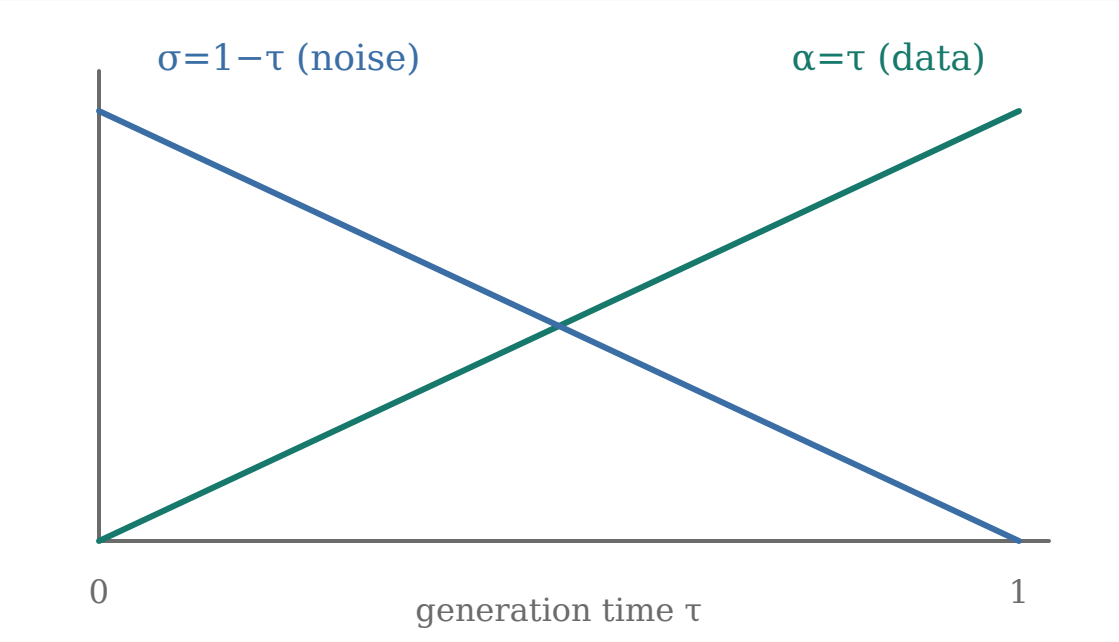
The Bridge to Diffusion

A Gaussian path can use different schedules

The conditional path adds scaled Gaussian noise to the data. The score is the gradient of the log marginal density.

$$X_\tau = \alpha(\tau)Y + \sigma(\tau)\varepsilon$$

where Y: data; $\varepsilon \sim \mathcal{N}(0, I)$: independent noise; X_τ: state at time τ; α, σ: smooth signal/noise scales, $\sigma > 0$.



Lipman et al., §4 and Appendix D; course score derivation

DERIVATION

The Bridge to Diffusion

Derive the score identity

$$X = \alpha Y + \sigma \varepsilon$$

$$g_y = \nabla \log p(x|y) = -(x - \alpha y)/\sigma^2$$

$$\nabla p = \mathbb{E}[p(x|Y) g_Y]$$

$$s = \nabla \log p = \mathbb{E}[g_Y|x] = -\mathbb{E}[\varepsilon|x]/\sigma$$

where Fixed τ. x, y: state/data; p: marginal density; g_y: conditional score. Scales α, σ, $\sigma > 0$; independent $\varepsilon \sim \mathcal{N}(0, I)$.

Lipman et al., §4 and Appendix D; course score derivation

DERIVATION

The Bridge to Diffusion

Score to velocity

$$X_\tau = \alpha Y + \sigma \varepsilon$$

$$v = \dot{\alpha} m + \dot{\sigma} n,\ n = -\sigma s$$

$$\alpha m = x - \sigma n$$

$$v = \dot{\alpha} x / \alpha + \sigma(\dot{\alpha}\sigma/\alpha - \dot{\sigma}) s$$

where x, v: state/rate; data Y; Gaussian noise ε; score s. $m = \mathbb{E}[Y|x], n = \mathbb{E}[\varepsilon|x]$. Scales $\alpha \neq 0, \sigma > 0$; dots: rates.

Lipman et al., §4 and Appendix D; course score derivation

DERIVATION

The Bridge to Diffusion

The straight-path bridge

$$v(x,\tau) = \frac{x}{\tau} + \frac{1-\tau}{\tau} s_\tau(x), \quad 0 < \tau < 1$$

$$\alpha = \tau, \quad \dot{\alpha} = 1, \quad \sigma = 1 - \tau, \quad \dot{\sigma} = -1$$

$$\frac{(1-\tau)^2}{\tau} + (1-\tau) = (1-\tau)/\tau$$

where x: state; s_τ: score of its density; v: velocity; dots: rates in τ.

Lipman et al., §4 and Appendix D; course score derivation

EXERCISE

The Bridge to Diffusion

Worked bridge: a known Gaussian path

Straight path $X_\tau = (1-\tau)\varepsilon + \tau Y$, fixed data $Y = 2$, standard Gaussian noise ε. At $\tau = 1/2$, $X_\tau \sim \mathcal{N}(1, 1/4)$ (variance $1/4$). Find score s and rate v at $x = 0$.

A s=4, v=4

B s=2, v=2

C s=−4, v=−4

Lipman et al., §4 and Appendix D; course score derivation

EXERCISE

The Bridge to Diffusion

Transfer: reverse the clock consistently

At course time $\tau = 1/2$ the rate is $+4$. Code time is $s = 1 - \tau$. For an Euler step $\Delta s = -.1$, find the code rate and state increment.

A Code velocity -4, state increment $+.4$

B Code velocity $+4$, state increment $-.4$

C Code velocity -4, state increment $-.4$

openpi 5546575: compute_loss / sample_actions

DEFINITION

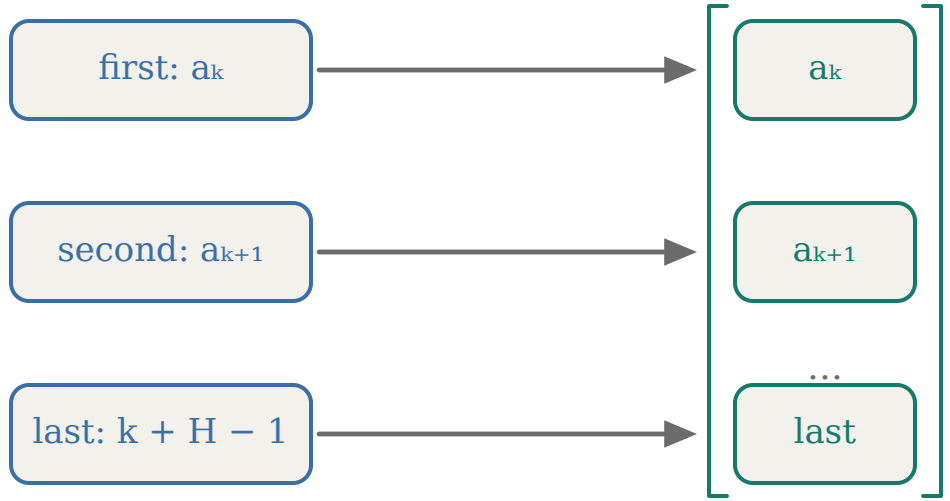
Vision-Language-Action Models

An action chunk is one generated vector

An action chunk stacks several successive action vectors into one generated vector.

$$A_k = \mathrm{vec}(a_k, \ldots, a_{k+H-1}) \in \mathbb{R}^{Hd_a}$$

where A_k: action chunk; k: robot index; H: chunk length; d_a: coordinates per action; vec: stack into a column.



Π0, §IV; Appendix A-B and A-D

DEFINITION

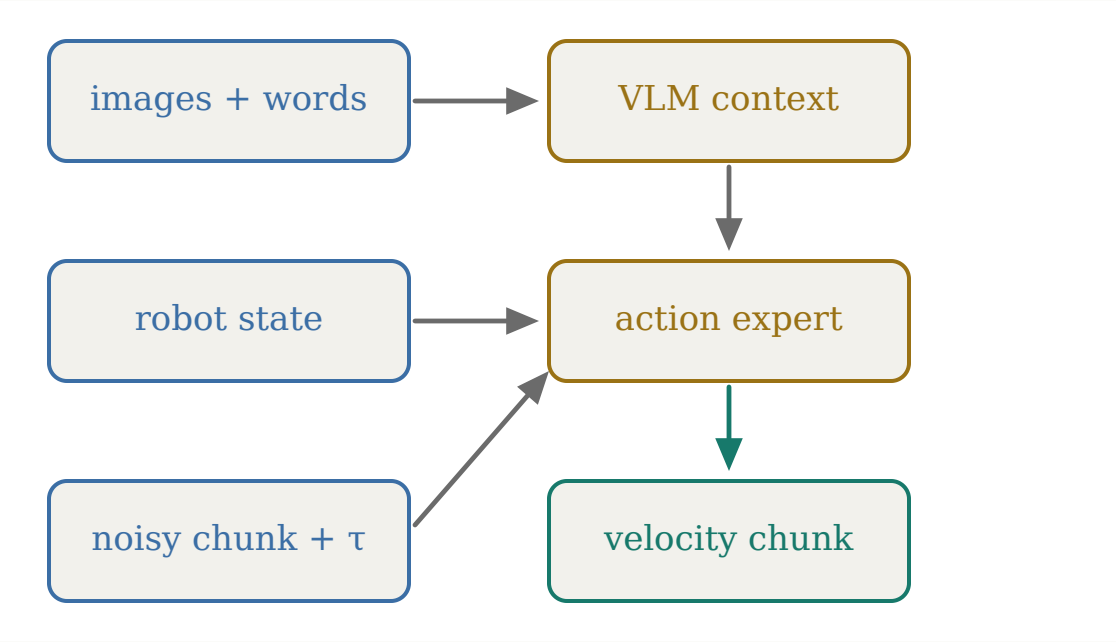
Vision-Language-Action Models

The condition describes the task now

A condition supplies the observed information held fixed during a sampling call.

$$c_k = (I_k^{1:C}, \ell_k, q_k), \qquad v_\theta(A_k^\tau, \tau, c_k)$$

where k: robot step; c_k: context; $I^{1:C}$: images from C cameras; ℓ : instruction; q: robot state; A^τ: candidate at generation time τ.



Π0, §IV; Appendix A-B and A-D

MODEL

Vision-Language-Action Models

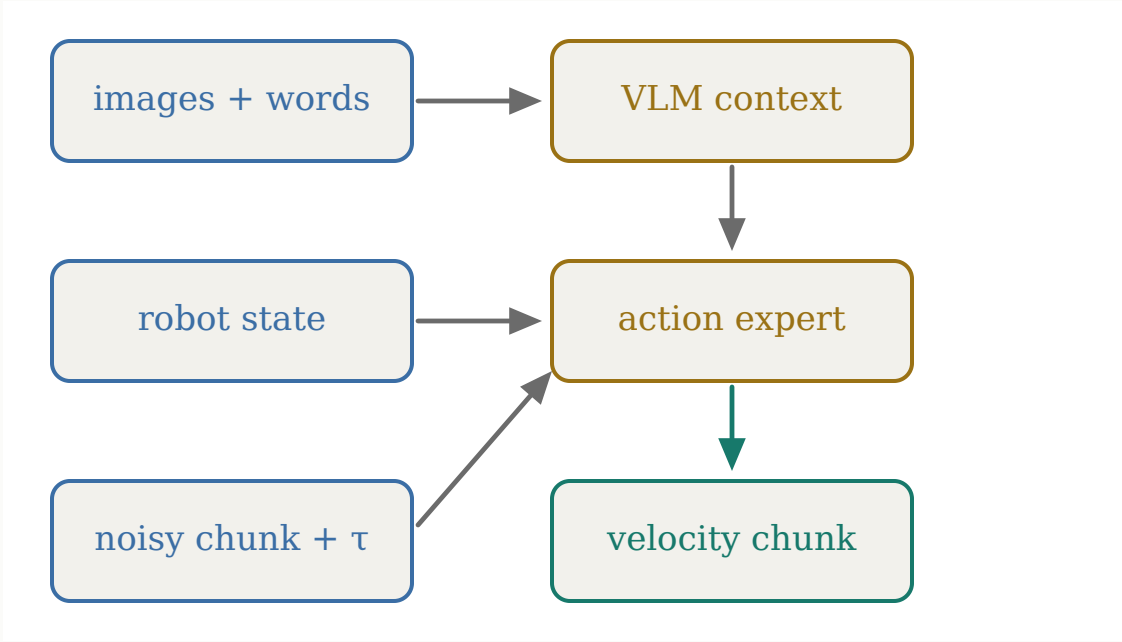
π0 adds a continuous action expert

An action expert is a separate set of Transformer weights specialized for action prediction.

VLM representations supply image-language context.

A smaller action expert processes noisy actions.

Attention connects context to action refinement.



π0, §IV; Appendix A-B and A-D

CONVENTION

Vision-Language-Action Models

Official code uses the opposite clock

$$Z_s = (1-s)A + s\varepsilon, \quad U_s = \varepsilon - A, \quad s : 1 \to 0$$

$$Z_s = X_{1-s}, \qquad \frac{dZ_s}{ds} = -\frac{dX_\tau}{d\tau}$$

$$\Delta s = -h, \quad U_s = -U_\tau \Rightarrow \Delta s\, U_s = hU_\tau$$

where A: data; ε: noise; Z_s, X_τ: code/course candidates; U_s, U_τ: rates; h: step.

openpi 5546575: Pi0.compute_loss / sample_actions

EXERCISE

Vision-Language-Action Models

A code step toward the data

Target $A = 2$, noise $\varepsilon = -1$. At code time $s = 1$, candidate $z = -1$. Find target rate u and the candidate after a step $\Delta s = -1/4$.

A u=−3, new z=−.25

B u=+3, new z=−1.75

C u=−3, new z=−1.75

openpi 5546575: Pi0.compute_loss / sample_actions

EXERCISE

Vision-Language-Action Models

Generation, execution, and replanning

Generate $H = 6$ actions. Execute the first $r = 2$, observe again, then generate a new chunk.

A. Six internal steps imply six robot actions.

B. Physical time advances by two executed actions.

C. The remaining four must always be executed.

π_0, §IV; Appendix A-B and A-D

MODEL

Vision-Language-Action Models

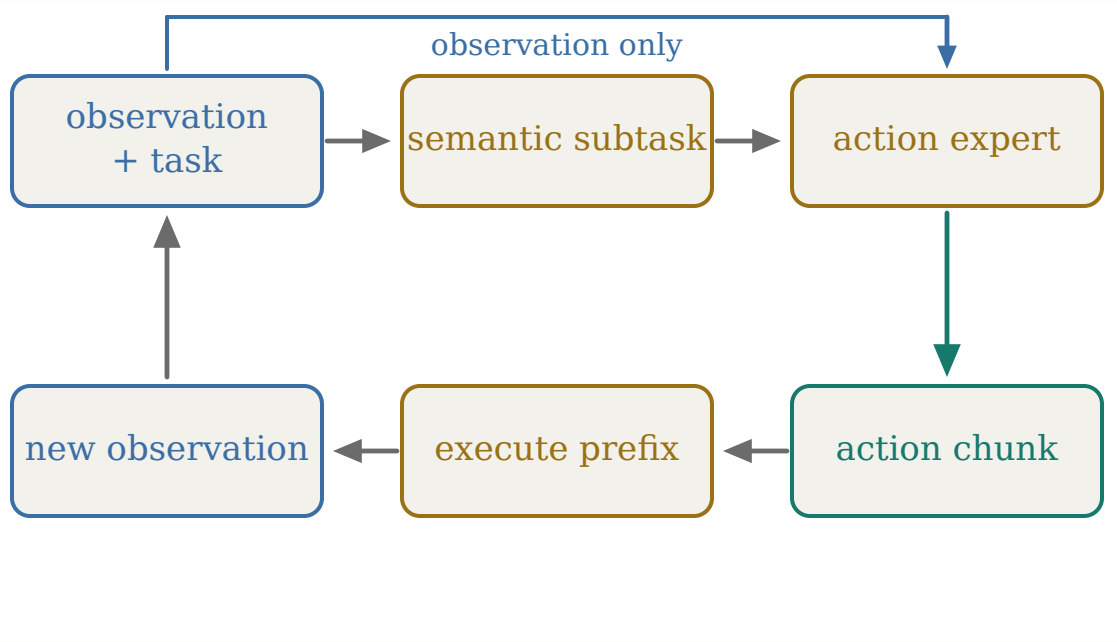
π0.5 connects semantic and action prediction

Semantic subtask prediction chooses a language description that conditions the low-level actions.

A high-level prediction identifies a subtask.

The action expert conditions on that subtask.

Training combines robot and non-robot sources.



π0.5, §§IV-A, IV-C–D

DEFINITION

Vision-Language-Action Models

Discrete and continuous losses

Token loss is the target token’s negative log probability.

$$L = L_{\text{token}} + \lambda L_{\text{flow}}$$

$$L_{\text{token}} = -\log p_\theta(y|c)$$

$$L_{\text{flow}} = \mathbb{E}\|v_\theta(Z_s, s, c) - U_s\|^2$$

where y: token; Z_s: noisy chunk; U_s: target rate; s: time; c: context; θ : weights; $\lambda > 0$: flow weight.

Po.5, §IV-B, Eq. (1); §IV-D

EXERCISE

Vision-Language-Action Models

What follows from a richer training mixture?

A vision-language-action model receives additional image-language and robot examples. Its low-level action model still uses flow matching.

A. The ODE becomes physical robot dynamics.

B. Representation learning changes; the flow objective can remain.

C. Generalization is guaranteed for every new home.

Π0.5, §§IV-C–D, V-C

EXAMPLE

From Data to an Executed Action

Train an action chunk

$$A = (2, 0)^T,\ \varepsilon = (-2, 2)^T,\ \tau = 1/4$$

$$A^\tau = .75\varepsilon + .25A = (-1, 1.5)^T$$

$$U = A - \varepsilon = (4, -2)^T, V = (3, -1)^T, e = V - U = (-1, 1)^T$$

$$L = (e_1^2 + e_2^2)/2 = 1,\ \nabla_V L = e$$

where A, ε: data/noise; τ: generation time; A^τ: noisy input; U, V: target/prediction; e: error; L: loss.

π_0, §IV and Appendix A-D; original course example

EXAMPLE

From Data to an Executed Action

Generate a chunk; execute a prefix

$$\varepsilon = (-2, 2)^T, v = (4, -2)^T, h = 1/4$$

$$x_j = \varepsilon + jhv$$

$$x_1 = (-1, 1.5)^T, x_2 = (0, 1)^T, x_3 = (1, .5)^T, x_4 = (2, 0)^T$$

$$H = 2, N = 4, r = 1 : \ a_k = 2$$

where x_j: step-j candidate; ε: noise; v, h: rate, step size. N, H, r: step count, chunk and prefix lengths. a_k: action at robot step k.

Π_0, §IV and Appendix A-D; original course example

EXERCISE

From Data to an Executed Action

Transfer: the observation changes mid-execution

After one executed action, a new image shows an obstacle. Old condition c described a clear route.

A. More ODE steps under old c incorporate the new image.

B. Update the condition and replan under the policy.

C. Low training MSE guarantees the old chunk is safe.

π_0, §IV and Appendix A-D; original course example

RECAP

From Data to an Executed Action

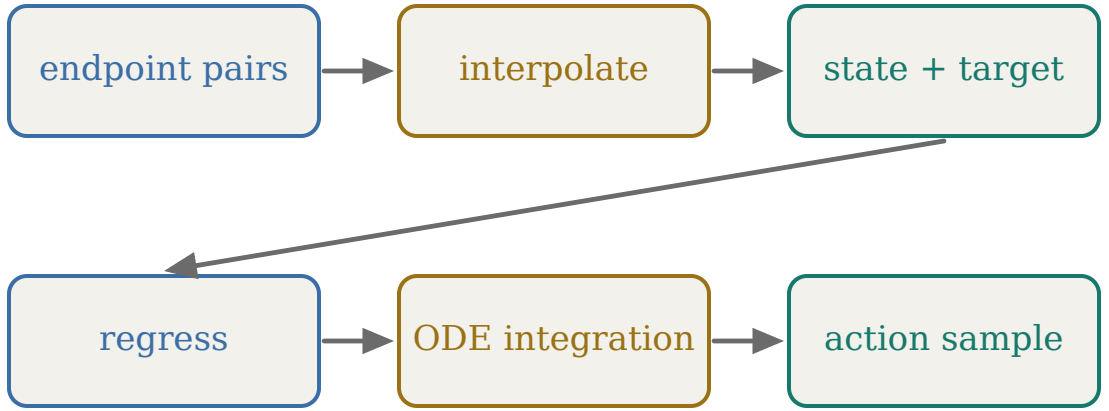
The course ingredients meet in one policy

Linear algebra represents chunks and attention.

Calculus trains a velocity predictor.

Probability defines the target law.

ODE integration generates a candidate action sequence.



fixed observation + language enter the regressor

Course synthesis; Tong et al., §3.1

References

Probability flows and regression

1. Lipman et al. Flow Matching for Generative Modeling. ICLR 2023, arXiv v2.
2. Tong et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport. TMLR 2024, arXiv v4.
3. Goodfellow, Bengio and Courville. Deep Learning. MIT Press, 2016, ch. 3.

References

Robot models and implementation conventions

1. Physical Intelligence. π_0: A Vision-Language-Action Flow Model for General Robot Control. arXiv v4, Jan 2026.
2. Physical Intelligence. $\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization. arXiv v1, Apr 2025.
3. openpi. Pi0.compute_loss / sample_actions. Commit 5546575, src/openpi/models/pi0.py.

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